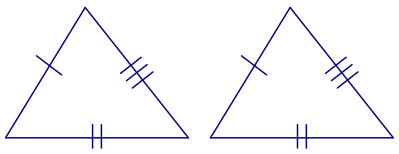
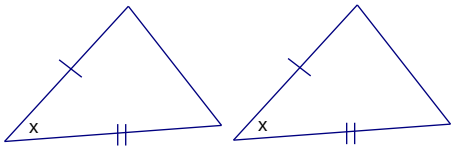
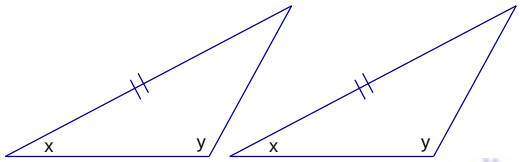
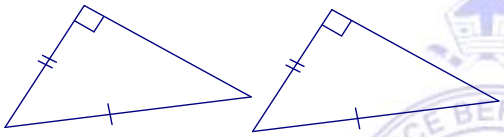
Grade 10 Congruency Questions

SUMMARY OF THE 4 CASES FOR CONGRUENCE OF TRIANGLES

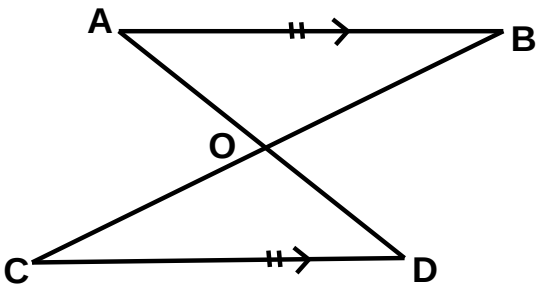
1. 	SSS = SSS
2. 	SAS = SAS (included angle)
3. 	AAS = AA corresponding S
4. 	RHS = RHS

Note: Triangles are ***not necessarily*** congruent if:

1. a a a = a a a
2. s s a = s s a For congruence the angle MUST BE the INCLUDED angle or a RIGHT ANGLE
3. a a s = a a s For congruence the sides MUST CORRESPOND, i.e. they must be opposite EQUAL ANGLES

QUESTION 1

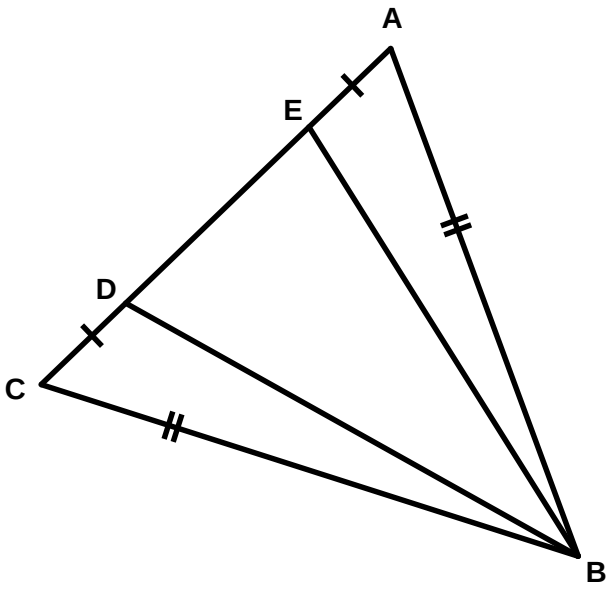
Prove that $AO = OD$ and $BO = OC$.



QUESTION 2

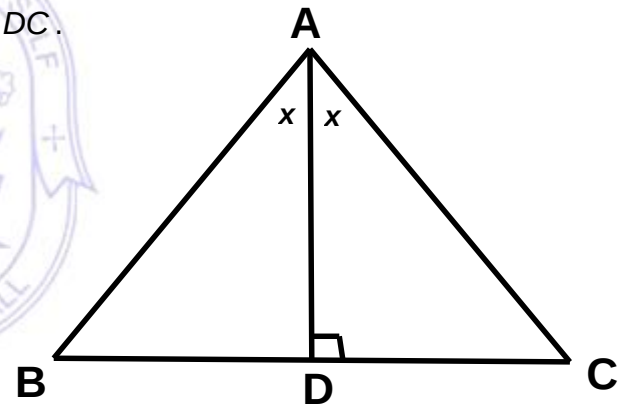
a) What type of triangle is ΔABC ?

b) Prove that $\Delta ABE \equiv \Delta CBD$.



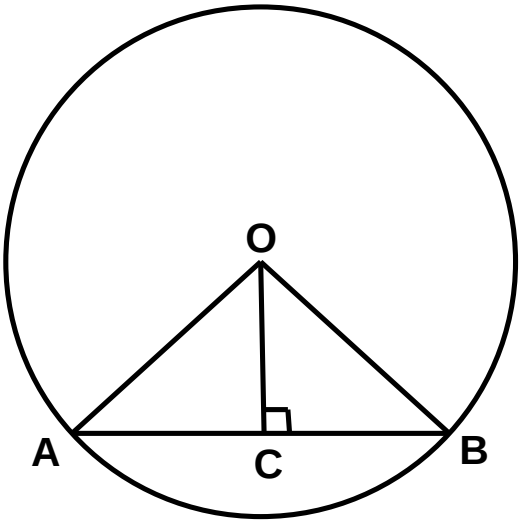
QUESTION 3

Prove that $AB = AC$ and $\hat{B} = \hat{C}$ and $BD = DC$.



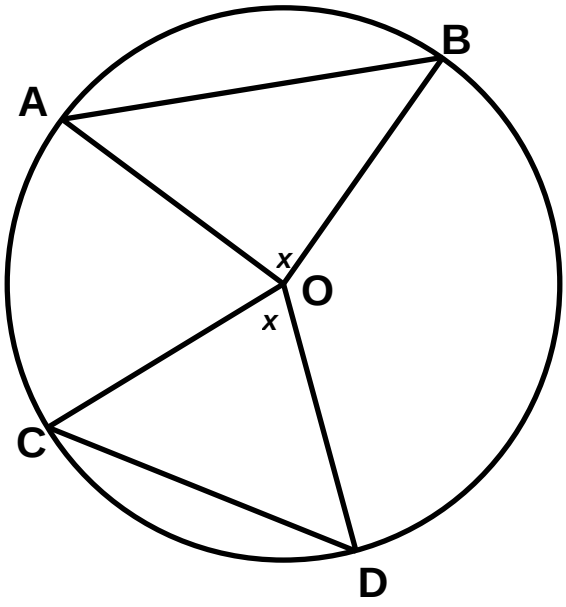
QUESTION 4

O is the centre of the circle. Prove that $AC = CB$.



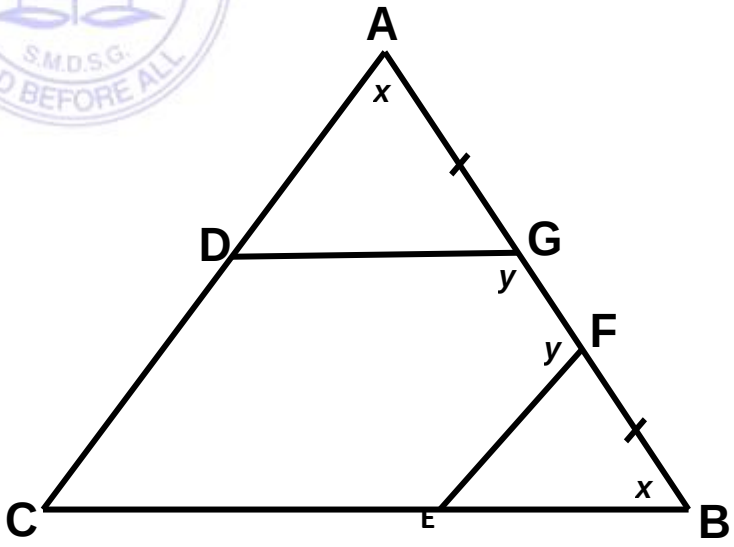
QUESTION 5

O is the centre of the circle. Prove that $AB = CD$.



QUESTION 6

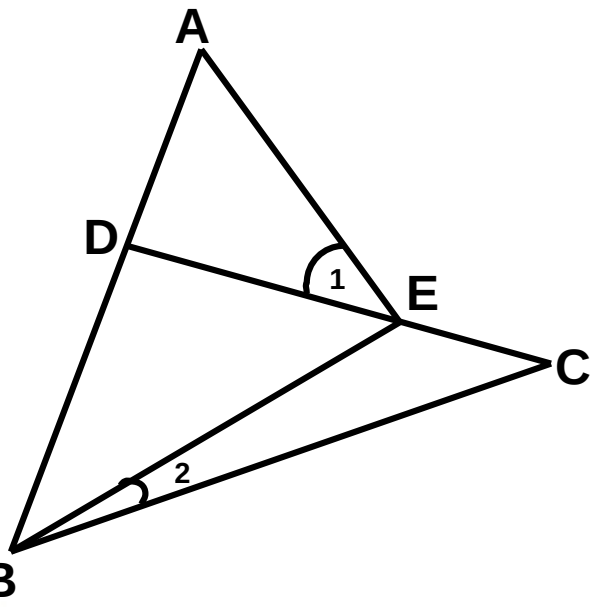
Prove that $AD = BE$.



QUESTION 7

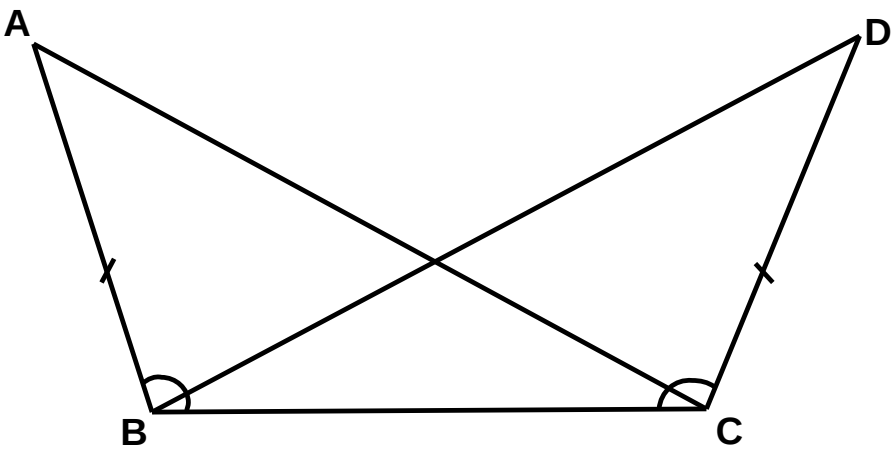
In the diagram ΔDBE is an equilateral triangle. $\hat{E}_1 = \hat{B}_2$
Prove that: a) $\Delta ABE \equiv \Delta CDB$.

b) $AD = EC$



QUESTION 8

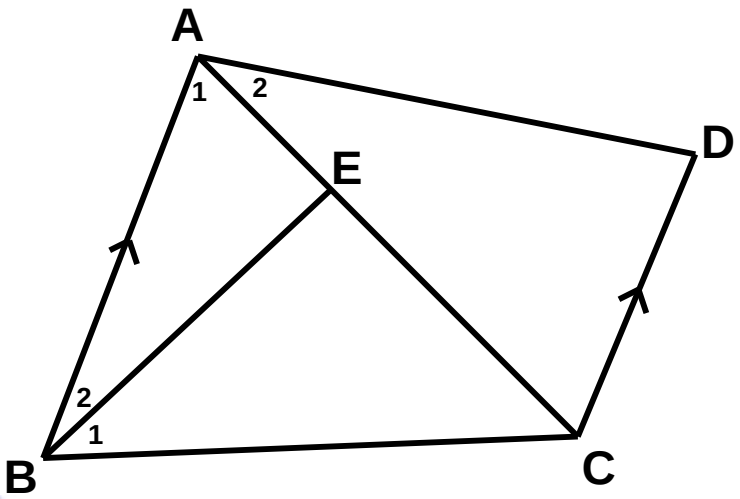
If $AB = DC$ and $A\hat{B}C = D\hat{C}B$, prove that $AC = DB$.



QUESTION 9

In the diagram: $AB = AC$ and $DC = AE$
Prove that:

a) $\Delta ACD \equiv \Delta BAE$

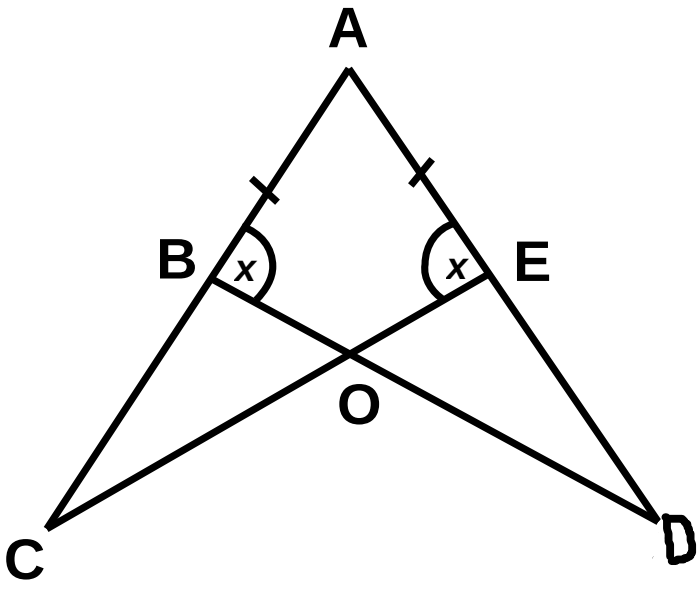


b) $B\hat{A}D = B\hat{E}C$

SERVICE BEFORE SELF
S.M.D.S.G.
GOD BEFORE ALL

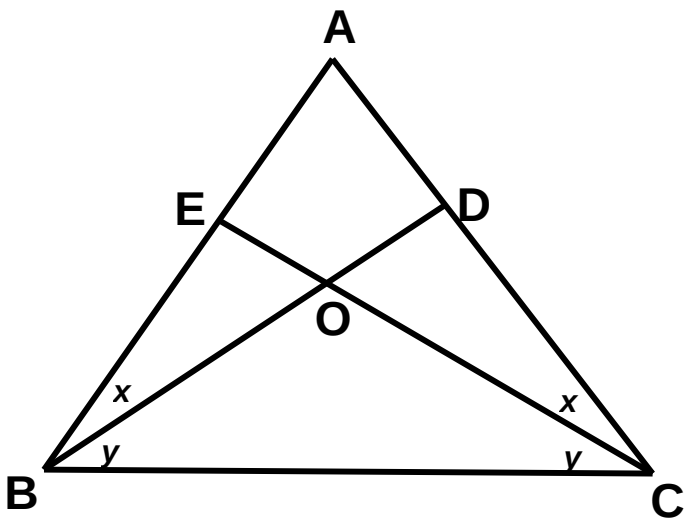
QUESTION 10

Given $AB = AE$. Prove that $CE = BD$.



QUESTION 11

Prove that $BD = CE$.



QUESTION 12***

FDCE is a parallelogram. CE is produced to A such that $CE = EA$ and $CD = DB$.
Prove that $\triangle BDF \equiv \triangle FEA$.

